

Euclidean Domains, PID, UFD

Vincent Wang

August 2025

1 Euclidean Domains

Let R be an integral domain and $N : R \rightarrow \mathbb{N}$ a multiplicative function such that $N(a) = 0$ iff $a = 0$, if $a, d \in R$ such that $d \neq 0$, then there exist $q, r \in R$ such that $a = dq + r$ and $N(r) < N(d)$.

Sps $N(a) = \begin{cases} 1 & \text{if } a \neq 0, \\ 0 & \text{if } a = 0. \end{cases}$ is a modulus. Let $d \neq 0$ be arbitrary and pick q, r st $1 = dq + r$. $N(r) < N(d) = 1 \rightarrow N(r) = 0 \rightarrow r = 0$, so R is a field. The GCD of any two elements is any nonzero element (unique up to units).

Suppose $a, b \in R$, both not zero. Define $\gcd(a, b)$ to be the set of positive N -minimal elements of $\langle a, b \rangle$ (the ideal generated by a, b).

$g \mid a, b$, and thus $g \mid \langle a, b \rangle$. Note $g \neq 0$. Let $q, r \in R$ such that $a = gq + r$, $N(r) < N(g)$. g is positive N -minimal, so $N(r) = 0 \rightarrow r = 0$.

If $d \mid a, b$, then $d \mid g$. $g = ax + by$, so $d \mid g$. If $a \mid b$ then $N(a) \mid N(b)$. $b = aq$, so $N(q)N(a) = N(b)$.

Suppose $u \in R^\times$. We have $N(1)^2 = N(1) \rightarrow N(1) = 0, 1$, but $1 \neq 0$. $N(1) = 1$. Pick $v \in R^\times$ such that $uv = 1 \rightarrow N(u)N(v) = 1 \rightarrow N(u) = N(v) = 1$.

Sps $N(a) = 1$. Then $a \in R^\times$. Note $a \neq 0$. Let $1 = aq + r$ for $N(r) < N(a) = 1 \rightarrow N(r) = 0 \rightarrow r = 0$.

If $N(a) = N(b)$, there does not necessarily exist $r \in R$ such that $a = br$. Consider $\mathbb{Z}[i]$ and note $|1 + 8i|^2 = |4 + 7i|^2$, $\frac{8+i}{4+7i} = \frac{3-4i}{5}$.

However, if $g \in \gcd(a, b)$ and u is a unit, then $gu \in \gcd(a, b)$.

$\mathbb{Z}[i]$ is indeed a Euclidean domain. $N(a + bi) = a^2 + b^2$ is zero iff $a + bi = 0$. N is multiplicative. If $a, d \in \mathbb{Z}[i]$ and d is nonzero, pick $q \in \mathbb{Z}[i]$ such that the integer and real parts are at most $1/2$ away from a/d . Then $N(a/d - q) \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

Then $N(a - dq) \leq \frac{1}{2}N(d) < N(d)$, where $r = a - dq \in \mathbb{Z}[i]$.

2 PID

Let $\mathcal{I}$ be a chain of proper ideals of R and $I = \bigcup \mathcal{I}$. If $a, b \in I$, let $a \in I_\alpha, b \in I_\beta$. Take $\gamma = \max(\alpha, \beta)$, so $a - b \in I_\gamma \subset I$. If $a \in I_\alpha$ and $r \in R$ is arbitrary, then $ar \in I_\alpha \subset I$. Similarly, $ra \in I$, so $I \triangleleft R$.

Sps FTSOC $I = R$. Then $1 \in I_\alpha$ for some α , so $I_\alpha = R$, cont.

Let R be a commutative ring with identity and M a maximal ideal. Then R/M has no proper ideals (by Lattice theorem). Let $a \in R/M$ be nonzero. Then $\langle a \rangle = R/M$, so there exists $b \in R/M$ such that $ab = \bar{1}$, so R/M is a field.

Conversely, if R/M is a field, then R/M has no proper ideals, so M is a maximal ideal.

Take $R = \mathbb{Z}[X]$. $xR \triangleleft R$. Claim: $R/xR \cong \mathbb{Z}$. Let the isomorphism be $F: R/xR \rightarrow \mathbb{Z}$ be the evaluation function at 0. If $f - g \in xR$, then $f(0) = g(0)$, so F is well-defined. $F(f + g) = f(0) + g(0) = F(f) + F(g)$. $F(fg) = f(0)g(0) = F(f)F(g)$. F is surjective. F is injective: If $F(f) = F(g)$, then $f(0) = g(0)$, so $(f - g)(0) = 0$, so $f - g \in xR$, so $f = g$ up to xR .

Suppose a ring R with $1 \neq 0$ has no proper ideals. Is R a field? Consider $\mathbb{H}$. Let $z_0 \in \mathbb{H}$ be nonzero fixed. Suppose $z \in \mathbb{H}$. Then $z = z \cdot \frac{\overline{z_0}}{|z_0|^2} \cdot z_0$, $\mathbb{H}$ has no proper ideals.

A PID is where every ideal is principal.